

Phase 1 study evaluating the safety, preliminary efficacy, and pharmacodynamics of recombinant interleukin-15 in combination with nivolumab and ipilimumab in patients with refractory cancers

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ABSTRACT

Purpose Combinations of immune checkpoint inhibitors and cytokine therapies have shown promise in cancer immunotherapy. We aimed to evaluate the safety, tolerability, and preliminary efficacy of recombinant human interleukin-15 (rhIL-15) in combination with nivolumab and ipilimumab in patients with advanced, refractory cancers.

Methods This open-label, non-randomized study employed a 3+3 dose-escalation design to determine the maximum tolerated dose (MTD) of rhIL-15 combined with fixed doses of nivolumab and ipilimumab. Safety and tolerability were assessed according to Common Terminology Criteria for Adverse Events (CTCAE) criteria, and preliminary efficacy was assessed according to Response Evaluation Criteria in Solid Tumors (RECIST) and immune-related RECIST (irRECIST) criteria. Pharmacodynamic studies evaluated changes in immune cell populations in peripheral blood and paired baseline and on-treatment tumor biopsies.

Results Thirty-one patients were enrolled, with a median age of 56 years (range: 24–81 years). Five patients received rhIL-15 with either nivolumab or ipilimumab in safety run-in cohorts, 26 patients received rhIL-15 with nivolumab plus ipilimumab. The most common cancer types were gastrointestinal (n=7) or gynecologic (n=5). The MTD was 1 µg/kg/day rhIL-15, 240 mg nivolumab, and 1 mg/kg ipilimumab. The triplet combination showed a manageable safety profile; the most common treatment-related adverse events (trAEs) were chills (20/26, 77%), fever (18/26, 69%), injection site reaction (15/26, 58%), anemia (14/26, 54%), and fatigue (14/26, 54%). Lymphopenia (4/26, 15%) was the most common grade 3/4 trAE. Pharmacodynamic analysis of tumor biopsies revealed increases in CD8+, CD8+CD3ζpY142+ (ie, activated CD8+T cells), PD-1+CD3+, and CD45RO+CD3+ (ie, memory T cells) in some patients. Tumor infiltration of natural killer (NK) and γδ T cells was also observed. NK cell populations in peripheral blood were modulated by

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Interleukin (IL)-15 is a cytokine that regulates T-cell responses and activates natural killer (NK) cells. It has been evaluated as monotherapy and in combination with immune checkpoint inhibitors (ICIs). However, pharmacodynamic responses induced by these combinations have not been fully elucidated.

WHAT THIS STUDY ADDS

⇒ The combination of rhIL-15 with nivolumab and ipilimumab was administered with an acceptable safety profile. rhIL-15 induced expansion of NK and γδ T cells in some peripheral blood samples. Infiltration of these cells into tumors was observed in biopsies, but the sample size was too small to correlate with clinical response. Mean CD4+cell count decreased significantly and mean CD8:CD4 T-cell ratio increased significantly in tumor biopsies, indicating consistent changes in some immune cell populations.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study demonstrates that rhIL-15 can be safely administered with nivolumab and ipilimumab. Studies focusing on tumor histologies known to be more responsive to ICIs may result in improved overall response rates. Pharmacodynamic responses measured here in immune cell populations in both tumor biopsies and peripheral blood can inform changes to dosing regimens in future studies.

rhIL-15. 11 of 26 patients (42%) had stable disease as a best response on the triplet regimen. A partial response was measured after cycle 1 in one patient (4%) with

cholangiocarcinoma and confirmed after cycle 2; this patient completed 16 cycles.

Conclusions The combination of rhIL-15 with nivolumab and ipilimumab was safe and induced changes in immune cell populations in some patients. However, preliminary signs of efficacy were limited. Pharmacodynamic findings may support further clinical development of this combination with alternative dosing regimens or combinations of rhIL-15 with other therapeutic modalities.

INTRODUCTION

Immune checkpoint inhibitors (ICIs) block an “off switch” for the immune system and facilitate the recognition and eradication of tumor cells. Nivolumab and ipilimumab, which target the programmed death-1 (PD-1) and cytotoxic T-lymphocyte-associated protein 4 pathways, respectively, were among the earliest ICIs to demonstrate clinical efficacy.^{1,2} They have since demonstrated efficacy across a range of cancer types.^{3–6} However, not all patients respond to these therapies, and resistance is a concern.^{7–9} Resistance can arise via intrinsic (ie, tumor) or extrinsic (eg, systemic or tumor microenvironment) mechanisms.⁸ As with traditional chemotherapies, one approach to overcoming resistance uses combination immunotherapy to stimulate the antitumor immune response through multiple different pathways.

Combinations of ICIs have demonstrated improved efficacy over their respective single agents and over standard-of-care chemotherapy regimens.^{10–13} The combination of ipilimumab plus nivolumab has been approved by the US Food and Drug Administration for treatment of a variety of cancer types including melanoma, renal cell carcinoma, non-small cell lung cancer, malignant pleural mesothelioma, hepatocellular carcinoma, esophageal squamous cell carcinoma, and microsatellite instability-high (MSI-H) or mismatch repair-deficient colorectal cancer.¹⁴ In a randomized phase 3 study (CheckMate 8HW, NCT04008030) comparing nivolumab plus ipilimumab to nivolumab alone in 707 participants with MSI-H metastatic colorectal cancer, progression-free survival (PFS) was significantly improved (HR 0.62, 95% CI 0.48 to 0.81, $p=0.0003$) in the combination arm over the nivolumab arm.¹⁵ Data cut-off occurred before median PFS in the combination arm was determined (95% CI 53.8 months to not estimable). In a long-term follow-up analysis of 945 randomized patients with advanced melanoma in the CheckMate 067 trial (NCT01844505), median overall survival (OS) was not reached in the nivolumab plus ipilimumab arm ($n=314$) at 120 months, whereas median OS survival in the nivolumab arm was 49.4 months ($n=316$) and 21.9 months in the ipilimumab arm ($n=315$).¹⁶ While PFS and OS were prolonged in the combination arms of these studies, overall response rates (ORRs) were 71% and 50% for CheckMate 8HW and CheckMate 067, respectively,^{15,16} leaving room for further improvement.

ICIs in combination with cytokine therapies have demonstrated efficacy in various cancer types.¹⁷ IL-15 is a critical cytokine in the regulation of natural killer (NK) and T-cell responses.¹⁸ rhIL-15 promotes the proliferation

and activation of these immune cells, leading to enhanced antitumor activity in clinical trials.^{18,19} The rationale behind combining rhIL-15 with nivolumab and ipilimumab lies in their complementary mechanisms of action.^{20,21} While nivolumab and ipilimumab release the inhibitory checkpoints on T cells, rhIL-15 can further enhance the proliferation and activation of these immune cells.^{18,19,22} In this context, the combination of rhIL-15 with nivolumab and ipilimumab presents a rational strategy to enhance the antitumor immune response.²³ We hypothesized that adding rhIL-15 to the combination of nivolumab with ipilimumab would amplify the antitumor immune response, potentially overcoming resistance to ICIs and improving treatment outcomes in patients with refractory cancers.^{23,24} Here, we report the results of a phase 1 study (ClinicalTrials.gov Identifier: NCT03388632) evaluating the safety, tolerability, and preliminary efficacy of this combination in patients with refractory cancers.

PATIENTS AND METHODS

Patient eligibility

Key inclusion criteria included being ≥ 18 years old with advanced, refractory malignancy that progressed, or the patient did not tolerate at least one prior systemic therapy, or for which no standard treatment exists. An Eastern Cooperative Oncology Group (ECOG) performance status of ≤ 2 (<https://ecog-acrin.org/resources/ecog-performance-status/>) and disease that was measurable according to RECIST V.1.1 criteria were also mandated.²⁵ Participants with autoimmune conditions (either treated within 2 years prior to enrollment or with any history of severe disease) and patients with active infections were excluded. Prior immunotherapy was allowed, but a 6-week washout period was required for any prior checkpoint inhibitor antibody before initiating study treatment.

Study design

This phase 1, open-label, non-randomized study was designed to evaluate the safety, MTD (primary objective), and preliminary efficacy (exploratory objective) of the combination of rhIL-15 with nivolumab and ipilimumab in patients with refractory cancers. The study was conducted at the National Institutes of Health Clinical Center (Bethesda, Maryland, USA) in accordance with the principles of the Declaration of Helsinki and Good Clinical Practice guidelines and sponsored by the National Cancer Institute (NCI) Division of Cancer Treatment and Diagnosis' (DCTD) Cancer Therapy Evaluation Program (CTEP). The protocol was approved by the NCI's Institutional Review Board (<https://www.hhs.gov/ohrp/index.html>). The protocol is available as an online supplemental appendix. Patients received subcutaneous (SC) rhIL-15 on days 1–8 and 22–29 of each 42-day cycle for the first four cycles (online supplemental figure S1). Nivolumab was administered (240 mg intravenous) on days 8, 22, and 36 of each cycle. Ipilimumab was administered (1 mg/kg intravenous) on day 8 of each of the first four cycles, then

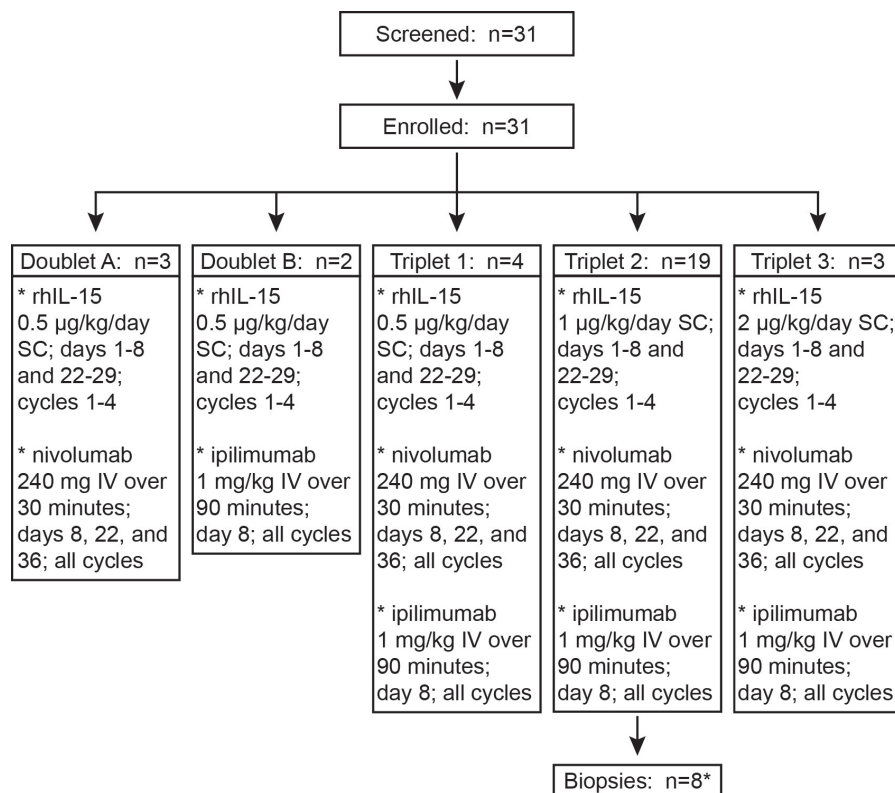


Figure 1 Flow diagram. 31 patients were screened and enrolled. Dose level assignments and dosing regimens are indicated. The Triplet 2 dose level was determined to be the MTD; paired pre-treatment and on-treatment biopsies were collected from patients in an expansion cohort at this dose level. IV, intravenous; MTD, maximum tolerated dose; rhIL-15, recombinant human interleukin-15; SC, subcutaneous.

on day 1 of each subsequent cycle. Dose reductions and interruptions were allowed. Treatment continued until disease progression, unacceptable toxicity, or withdrawal of consent. All patients who received at least one dose of study drug were considered to be evaluable for response. Nivolumab and ipilimumab were provided to the NCI/DCTD through Cooperative Research and Development Agreements; rhIL-15 was supplied by the NCI Pharmaceutical Management Branch.

Lead-in doublet groups of up to three patients each were enrolled to evaluate rhIL-15 with either nivolumab (Doublet A) or ipilimumab (Doublet B, [figure 1](#)); these patients were assigned to either group at the investigator's discretion. A doublet was deemed safe if two of three patients in that doublet completed cycle 1 without experiencing a dose-limiting toxicity (DLT); DLT criteria are listed in the online supplemental methods. Advancing from the doublet to the triplet part of the study required review and approval from CTEP and the Institutional Review Board per-protocol. The triplet part of the study used a 3+3 dose-escalation design to determine the MTD of rhIL-15 when administered in combination with fixed doses of nivolumab and ipilimumab as described above. Patients were enrolled in cohorts of three to six patients, with dose escalation proceeding based on the assessment of DLTs during the first cycle of treatment (42 days). The dose-escalation plan included up to three dose levels of rhIL-15 (0.5, 1, and 2 µg/kg/day on days 1–8 and 22–29

of cycles 1–4). Once the MTD was established, additional patients were enrolled in an expansion cohort to further evaluate the safety and efficacy of the combination at that dose level. The expansion phase plan included up to 15 patients, with each patient receiving up to four cycles of rhIL-15.

Safety assessments

Safety assessments included monitoring for adverse events (AEs), serious AEs, DLTs, and changes in laboratory parameters, vital signs, and physical examination findings. AEs were graded according to the CTCAE V.4.0 (on or before March 31, 2018) or V.5.0 (on or after April 1, 2018) (https://ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm). All patients who received at least one dose of study drug were evaluable for safety and toxicity.

Tumor measurements

Tumor response was assessed using the RECIST V.1.1²⁵ and iRECIST criteria for immunotherapeutic studies.²⁶ Imaging studies were performed before the start of every cycle (approximately every 6 weeks) for the first four cycles to evaluate treatment response and every other cycle thereafter.

Immunofluorescence microscopy

Paired pre-treatment and on-treatment samples were collected to assess changes in the tumor microenvironment

(TME). Tumor biopsies were flash-frozen within 2 min from the time of collection to preserve labile biomarkers (eg, phosphorylation) according to NCI DCTD Standard Operating Procedures (SOP340507, <https://dctd.cancer.gov/ResearchResources/biomarkers/sample-collection.htm>).²⁷ Immunohistochemistry (IHC) and multiplex immunofluorescence were used to evaluate the infiltration and activation status of immune cells within the tumor tissue. Immune cell populations including cytotoxic T lymphocytes (CTLs, CD8+), activated CTLs (CD8+CD3 ζ pY142+), PD-1+CD3+ cells, programmed death-ligand 1 cells (PD-L1)+, memory T cells (CD45RO+CD3+), T helper cells (CD4+), and regulatory T cells (T_{reg} cells, CD4+FoxP3+) were analyzed largely as previously described²⁸ except that β -catenin (Clone E247) was used as a tumor segmentation marker for the carcinoma specimens. The entire tumor area and stroma within 500 μ m of tumor were analyzed using Indica Labs' HALO and HALO AI software employing nuclear segmentation (using the 4',6-diamidino-2-phenylindole (DAPI) layer) and custom marker thresholding to determine cellular densities (cells/mm²). Samples that did not reach a minimum of 3,000 cells in the analyzable area were not reported. Mean values of pre-treatment and on-treatment biopsies were compared using two-sided Wilcoxon matched-pairs signed-rank test.

TCAP-NK flow cytometry

Human whole blood samples were collected and fixed using BD Phosflow Lyse/Fix Buffer (BD Biosciences) on-site. Following fixation, cells were washed and resuspended in a buffer composed of 1 $\times$ phosphate-buffered saline supplemented with 60% (v/v) methanol, and 0.25 mM 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid (HEPES), then snap-frozen at -80°C . The frozen samples were transported to the Frederick National Laboratory for Cancer Research for batch flow cytometry acquisition. For T Cell Activation Panel (TCAP)-NK flow cytometry staining, samples were thawed, washed with Perm/Wash Buffer (BD Biosciences), and subsequently mixed with antibodies from the TCAP-NK panel (online supplemental table S1). Fluorescence minus one (FMO) controls were established by staining cells from pre-dose samples with specific antibody sets corresponding to the FMO controls. Immediately after staining, the cells were analyzed using a BD LSRFortessa flow cytometer (BD Biosciences). The resulting datasets were processed with FlowJo V.10 (FlowJo). Cell population subsets, including NK cells and $\gamma\delta$ T cells, were visualized using Prism V.10 (GraphPad Software).

Statistical analysis

Once the MTD was established, up to 15 additional patients were to be enrolled in the expansion phase of the trial with mandatory tumor biopsies for assessment of pharmacodynamic endpoints. With up to 15 patients and a tumor biopsy quality assurance criteria failure rate of 50% with respect to paired (pre-dose and post-dose)

biopsies, we had an 85% likelihood of having at least six usable biopsy pairs, and 95% likelihood of having at least five usable samples for analysis. With the analysis of six biopsy pairs, there would be 90% power to detect a treatment effect of 1.85 SDs, based on the intrapatient variability of each specific marker, using a two-sided Wilcoxon matched-pairs signed-rank test at the 0.05 significance level.

RESULTS

Study population

A total of 31 patients were enrolled in this trial between February 6, 2018, and July 16, 2024 (table 1). The median age of the patients was 56 years (range: 24–81 years). Several different cancer types were represented in the study population (table 1, online supplemental table S2). A majority of patients (21/31, 68%) had an ECOG performance status of 1, nine patients (29%) had a performance status of 0, and one patient (3%) had a performance status of 2 (table 1). Patients were heavily pretreated, with a median of 3 prior therapies (range: 2–5), including systemic chemotherapy, surgery, radiation, immunotherapy, and/or hormone therapy. Patients had received a median of 2 prior lines of systemic therapy (range: 0–3, including chemotherapy and hormone therapy); two patients received no prior systemic therapy. Notably, four patients (13%, patients 5, 15, 27, and 28) had received prior immunotherapy. Accrual was halted after the MTD was determined and sufficient analyzable biopsies had been collected to complete exploratory objectives. All patients completed the study intervention on or before November 27, 2024, and data were cut-off on December 9, 2024.

MTD determination

No DLTs occurred in either of the doublet cohorts (online supplemental table S3). At the Triplet 2 dose level, a patient experienced prolonged grade 2 increased lipase that was considered dose-limiting. This dose level was expanded to 6 patients; 10 patients were enrolled at Triplet 2 during dose escalation, but 4 patients were not evaluable for toxicity per protocol (ie, did not experience a DLT but did not complete cycle 1) and were replaced for a total of 6 evaluable patients. No additional DLTs occurred. Three patients were enrolled at the Triplet 3 dose level; one patient experienced grade 3 photosensitivity and another patient experienced grade 2 dermatitis that was identified as a serum sickness-like reaction requiring a 6-week course of antihistamine treatment. The Triplet 2 dose level (1 μ g/kg/day SC rhIL-15) was determined to be the MTD, and nine additional patients were enrolled in an expansion cohort at this dose level.

Adverse events

All 31 patients enrolled in the study were evaluable for toxicity. trAEs were observed in all patients. trAEs occurring in patients receiving a doublet regimen (n=5) are

Table 1 Baseline patient demographics

Characteristic	Patients, n (%)
Sex	
Female	18 (58)
Male	13 (42)
Age, median years (range)	56 (24–81)
ECOG performance status	
0	9 (29)
1	21 (68)
2	1 (3)
Ethnicity	
Non-Hispanic or Latino	28 (90)
Hispanic or Latino	3 (10)
Race	
White	21 (68)
Black	6 (19)
Asian	3 (10)
Not reported	1 (3)
Primary cancer	
Gastrointestinal*	7
Gynecologic†	5
Sarcoma‡	5
Hepatobiliary	4
Pancreas	4
Breast	2
Thyroid	2
Mesothelioma	1
Other (parotid carcinoma)	1
Total prior therapies	
2	8 (26)
3	8 (26)
4	14 (45)
5	1 (3)
Prior systemic therapy§	
0	2 (6)
1	10 (32)
2	17 (55)
3	2 (6)
Prior surgery	
Yes	31 (100)
No	0 (0)
Prior radiation	
Yes	19 (61)
No	12 (39)
Prior immunotherapy	
Yes¶	4 (13)
No	27 (87)

Continued

Table 1 Continued

Characteristic	Patients, n (%)
*Includes colon (n=4), rectum (n=1), stomach (n=1), and gastroesophageal junction (n=1).	
†Includes uterus (n=3), cervix (n=1), and ovary (n=1).	
‡Includes: includes synovial sarcoma, cardiac intimal sarcoma, pleiomorphic sarcoma, alveolar soft part sarcoma, and osteosarcoma (n=1 each).	
§Number of prior regimens (either monotherapy or combination chemotherapy regimens or hormone therapy).	
¶Includes immunotherapy (n=3) and cancer vaccine therapy (n=1).	
ECOG, Eastern Cooperative Oncology Group.	

listed in online supplemental table S3. Among patients in the triplet cohort (n=26), chills (20/26, 77%), fever (18/26, 69%), injection site reaction (15/26, 58%), anemia (14/26, 54%), and fatigue (14/26, 54%) were the most prevalent trAEs (table 2). Lymphopenia (4/26, 15%) was the most common grade 3/4 trAE. One occurrence of lymphopenia was grade 4, while three patients experienced grade 3 lymphopenia; no other grade 4 trAEs occurred. Other grade 3 trAEs included neutropenia (n=2) and one instance each of anemia, colitis, decreased white blood cell count, dehydration, diarrhea, and hyponatremia. Patient 22 (Triplet 3) had a dose reduction during cycle 1 for the serum sickness-like reaction mentioned above and completed three cycles. Dose interruptions occurred in patients at the Doublet A (2 of 3 patients, 67%), Triplet 1 (2 of 4 patients, 50%), Triplet 2 (13 of 19, 68%), and Triplet 3 (3 of 3 patients, 100%) dose levels. However, none of the trAEs that led to the dose reductions met the definitions of DLTs. Three patients (10%) discontinued treatment due to trAEs for colitis (patients 5 and 10) and increased aspartate aminotransferase (AST) and lipase requiring hospitalization (patient 31).

Only one occurrence of treatment-related pneumonitis (patient 7, parotid carcinoma) was recorded during this study. Grade 1 pneumonitis occurred during cycle 2 and resolved during cycle 3 without intervention; this patient went on to complete 10 cycles of treatment. Patient 17 experienced grade 3 myocarditis near the end of cycle 2. Their disease progressed at the end of that cycle, and they discontinued study treatment as a result. The myocarditis resolved during the follow-up period following treatment with corticosteroids. Colitis occurred in two patients (patient 5 on Doublet A and patient 10 on Triplet 2, both grade 3). Patient 5 (rectal adenocarcinoma) was hospitalized for colitis during cycle 2, and patient 10 experienced colitis during cycle 5 (ie, after cessation of rhIL-15). Both were treated with corticosteroids and taken off study for toxicity.

Clinical outcomes

Thirty-one patients were evaluable for response according to RECIST V.1.1 criteria,²⁵ 5 patients in the

Table 2 Treatment-related adverse events in the triplet cohort (n=26)

Adverse event	Total, n (%)	Grade			
		1	2	3	4
Chills	20 (77)	20			
Fever	18 (69)	14	4		
Injection site reaction	15 (58)	13	2		
Anemia	14 (54)	6	7	1	
Fatigue	14 (54)	12	2		
Aspartate aminotransferase increased	9 (35)	8	1		
Lymphocyte count decreased	7 (27)	1	2	3	1
White blood cell count decreased	7 (27)	2	4	1	
Alanine aminotransferase increased	7 (27)	6	1		
Anorexia	7 (27)	6	1		
Nausea	7 (27)	7			
Neutrophil count decreased	6 (23)	1	3	2	
Diarrhea	6 (23)	5		1	
Platelet count decreased	5 (19)	4	1		
Headache	5 (19)	5			
Hyponatremia	4 (15)	3		1	
Alkaline phosphatase increased	4 (15)	3	1		
Hyperhidrosis	4 (15)	3	1		
Protein/creatinine ratio increased	4 (15)	4			
Serum amylase increased	4 (15)	4			
TSH increased	4 (15)	4			
Eczema	3 (12)		3		
Hypothyroidism	3 (12)		3		
Lipase increased	3 (12)		3		
Blood bilirubin increased	3 (12)	2	1		
Creatinine increased	3 (12)	2	1		
Dry mouth	3 (12)	2	1		
Cough	3 (12)	3			
Hyperthyroidism	3 (12)	3			
Lactate dehydrogenase increased	3 (12)	3			
Malaise	3 (12)	3			
Vomiting	3 (12)	3			
Adrenal insufficiency	2 (8)		2		
Hypertension	2 (8)		2		
Dry skin	2 (8)	1	1		
Hypokalemia	2 (8)	1	1		
Hypophosphatemia	2 (8)	1	1		
Insomnia	2 (8)	1	1		
Proteinuria	2 (8)	1	1		
Rash—maculopapular	2 (8)	1	1		

Continued

Table 2 Continued

Adverse event	Total, n (%)	Grade			
		1	2	3	4
Dizziness	2 (8)	2			
Hypotension	2 (8)	2			
Infusion-related reaction	2 (8)	2			
Colitis	1 (4)			1	
Dehydration	1 (4)			1	
Dysgeusia	1 (4)		1		
Hypoalbuminemia	1 (4)		1		
Hypophysitis	1 (4)		1		
Lymphocyte count increased	1 (4)		1		
Pain	1 (4)		1		
Photosensitivity	1 (4)		1		

No grade 5 trAEs occurred during this study. One instance each of the following trAEs occurred at grade 1: arthralgia, back pain, cardiac troponin I increased, conjunctivitis, dermatitis—serum sickness-like reaction, dyspnea, ear pain, edema—limbs, erythematous macule, influenza-like symptoms, GGT increased, hemoglobinuria, hot flashes, hyperglycemia, hypermagnesemia, hyperphosphatemia, mucositis—oral, myalgia, pneumonitis, sinus tachycardia, and weight loss.

GGT, gamma-glutamyl transferase; trAE, treatment-related adverse event; TSH, thyroid-stimulating hormone.

doublet cohorts and 26 patients in the triplet cohort. Among the five patients receiving a doublet regimen, three patients had stable disease (SD) and two patients had progressive disease as best responses; these patients completed a median of 3 cycles (range: 1–5 cycles). Twenty-six patients received the triplet regimen and were evaluable for response. These patients completed a median of 1 cycle of treatment (range: 0–16 cycles, [figure 2](#)). A confirmed partial response (PR) was measured in one patient (patient 8, cholangiocarcinoma, Triplet 1) for an ORR of 4% for the triplet combination. The initial PR was measured after completing cycle 1, confirmed after cycle 2, and continued through the end of cycle 16 when the patient chose to begin an alternative treatment. SD was measured in 11 of 26 patients (42%, median: 2 cycles, range: 1–4 cycles). Ten patients experienced progressive disease as best response, 2 additional patients (patients 11 and 30) had clinical progression during cycle 1 prior to their first re-staging scan. Two additional patients did not have tumor measurements after cycle 1, as one patient chose to discontinue study treatment and the other patient was removed following a 42-day lapse in treatment. Notably, the two longest courses of study treatment (patients 7 and 8) both occurred in patients at the Triplet 1 dose level (ie, the lowest dose of rhIL-15, online supplemental figure S2); the two largest per cent decreases in tumor measurements (patients 6 and 8) also occurred in patients at the Triplet 1 dose level.

Responses were also evaluated using iRECIST criteria.²⁶ 10 patients were treated beyond PD under iRECIST: 2

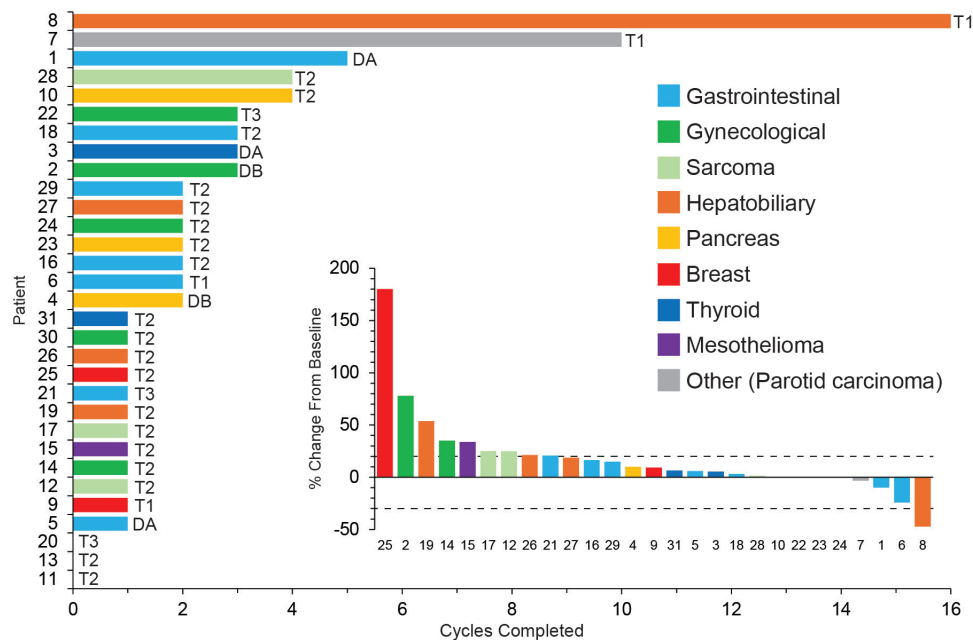


Figure 2 Cycles completed and best responses by tumor type. Bars are colored by tumor type as indicated. Dose levels for each patient are indicated. A confirmed PR was measured in one patient (patient 8, cholangiocarcinoma, Triplet 1) for an ORR of 3%. SD was measured in 17 of 31 patients (55%) and progressive disease was the best response in 9 patients (29%). DA, Doublet A; DB, Doublet B; ORR, overall response rate; PR, partial response; SD, stable disease; T1, Triplet 1; T2, Triplet 2; T3, Triplet 3.

patients on Doublet A, 2 patients on Doublet B, 1 patient on Triplet 1, 4 patients on Triplet 2, and 1 patient on Triplet 3 (online supplemental table S4). Patient 7 experienced unconfirmed progressive disease (iUPD) after cycle 1 with a 27.7% increase in tumor size from baseline. This was followed by tumor regression to 25.3% below maximum measurement after cycle 6, then confirmed progression after cycle 10 (21.0% increase from nadir). All other patients treated under iRECIST chose to come off study or had confirmed progressive disease (iCPD).

Immune cell populations in peripheral blood

Populations of NK cells and $\gamma\delta$ T cells in peripheral blood samples from six patients (patients 26–31) were analyzed longitudinally by flow cytometry. Antibodies were obtained from commercial sources, and validation information is provided on the manufacturer's websites (online supplemental table S1). During the first series of rhIL-15 injections on days 1–8 of cycle 1, NK cell populations decreased from baseline then expanded in five of six patients (patients 26, 28, 29, 30, and 31; figure 3A); these increases occurred prior to administration of nivolumab or ipilimumab. Subsequent decreases were measured in all five patients following the break in rhIL-15 administration from C1D9 to C1D21. rhIL-15 administration resumed on C1D22, and blood was not collected until C1D36 (ie, 7 days after the previous dose of rhIL-15). Another increase in NK cell population was measured in blood from patient 28 (alveolar soft part sarcoma (ASPS)) from C2D1 to C2D8. No such increase was measured after further series of rhIL-15 administration on C3D8 and C3D28. Fewer samples were analyzed

during cycle 2 and beyond as patients 26 and 30 came off the study after cycle 1. Increases in $\gamma\delta$ Vd1+ and $\gamma\delta$ Vd2+T cells were measured in peripheral blood, most notably for patients 26 and 28 (figure 3B,C, respectively). Transient decreases in both cell populations were measured in blood from patient 28 between C2D1 and C2D2. Populations of these cells were consistently lower during cycle 3 than in cycle 2, although blood was collected only from patient 28 during cycle 3. Infiltration of NK and $\gamma\delta$ T cells was evaluated in biopsies from patients 23, 26–29, and 31 and increases in circulating NK and $\gamma\delta$ T cells did not consistently translate into increased infiltration into tumors. Increased NK cell infiltration was observed in the C1D42 biopsy from patient 28 (figure 3D) and increased $\gamma\delta$ T-cell infiltration was observed in the C1D42 biopsy from patient 26 (figure 3E). Levels of NK and $\gamma\delta$ T cells were low or absent in most biopsies evaluated by IHC, indicating that increases in these cell populations in blood did not lead to increased density of these cells in tumors on C1D42 for most patients.

Immune cell infiltration in tumor biopsies

Paired biopsies were collected from eight patients at the Triplet 2 dose level on days 1 and 42 of cycle 1. Six of the pairs included analyzable tissue of sufficient quality to yield quantifiable data (patients 23, 26, 27, 28, 29, and 31). The six patients from whom these biopsies were collected completed a median of 2 cycles (range: 1–4 cycles). No objective responses were measured; best responses included SD (n=5) and progressive disease (n=1). Histologies included pancreatic cancer, cholangiocarcinoma, gallbladder cancer, ASPS,

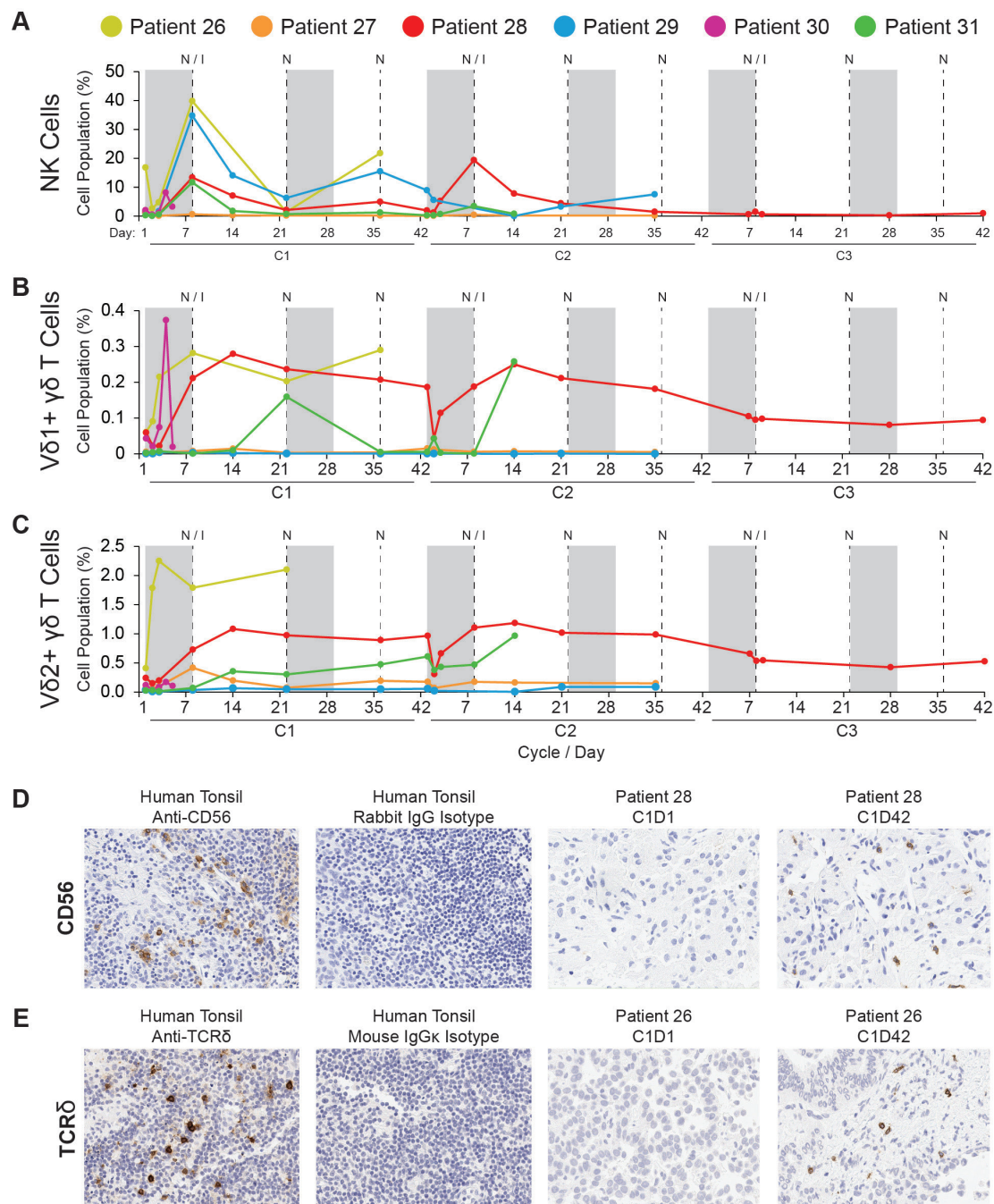


Figure 3 NK and γ/δ T-cell dynamics during treatment. Changes in individual patient cell populations from cycle 1 to cycle 3 are illustrated for patients 26, 27, 28, 29, 30, and 31 as indicated. Peripheral blood data are presented for NK cells (A), Vδ1+ γ/δ T cells (B), and Vδ2+ γ/δ T cells (C) and IHC of biopsy tissue for NK cells (D) and γ/δ T cells (E). Cell population percentages were calculated by dividing the number of CD56+ cells (ie, NK cells), Vδ1+ cells (for Vδ1+ γ/δ T cells), and Vδ2+ cells (for Vδ2+ γ/δ T cells) by the total cell population as defined in the forward scatter/side scatter dot plot (online supplemental figure S3). IHC, immunohistochemistry; NK, natural killer; TCR, T cell receptor.

rectal adenocarcinoma, and Hürthle cell carcinoma (all n=1). Tumor and stroma were segmented in biopsy images from patients 23, 26, 27, 29, and 31 using β -catenin as a carcinoma tumor marker. Biopsy images from patient 28 (ASPS) were segmented using TFE3 for evaluation of CD8+, CD8+CD3 ζ pY142+, and PD-L1+ cells; for assessment of the other immune cell subsets from this patient, tumor plus stroma within 500 μ m of the tumor were reported. Mean CD4+ cell population decreased

significantly ($p=0.0312$) and mean CD8+:CD4+ ratio increased significantly ($p=0.0312$). No other statistically significant changes in mean cells/mm² values and T_{reg}:CD8 ratio from C1D1 to C1D42 were measured in this biopsy cohort ($p \geq 0.0625$, figure 4). CD8+ T cells were observed in both tumor and stroma of all biopsies at both time points, with increases in CD8+ T cells measured in 4/6 patients (patients 23, 26, 28, and 31). Increases in the populations of CD45RO+CD3+ T cells (ie, memory

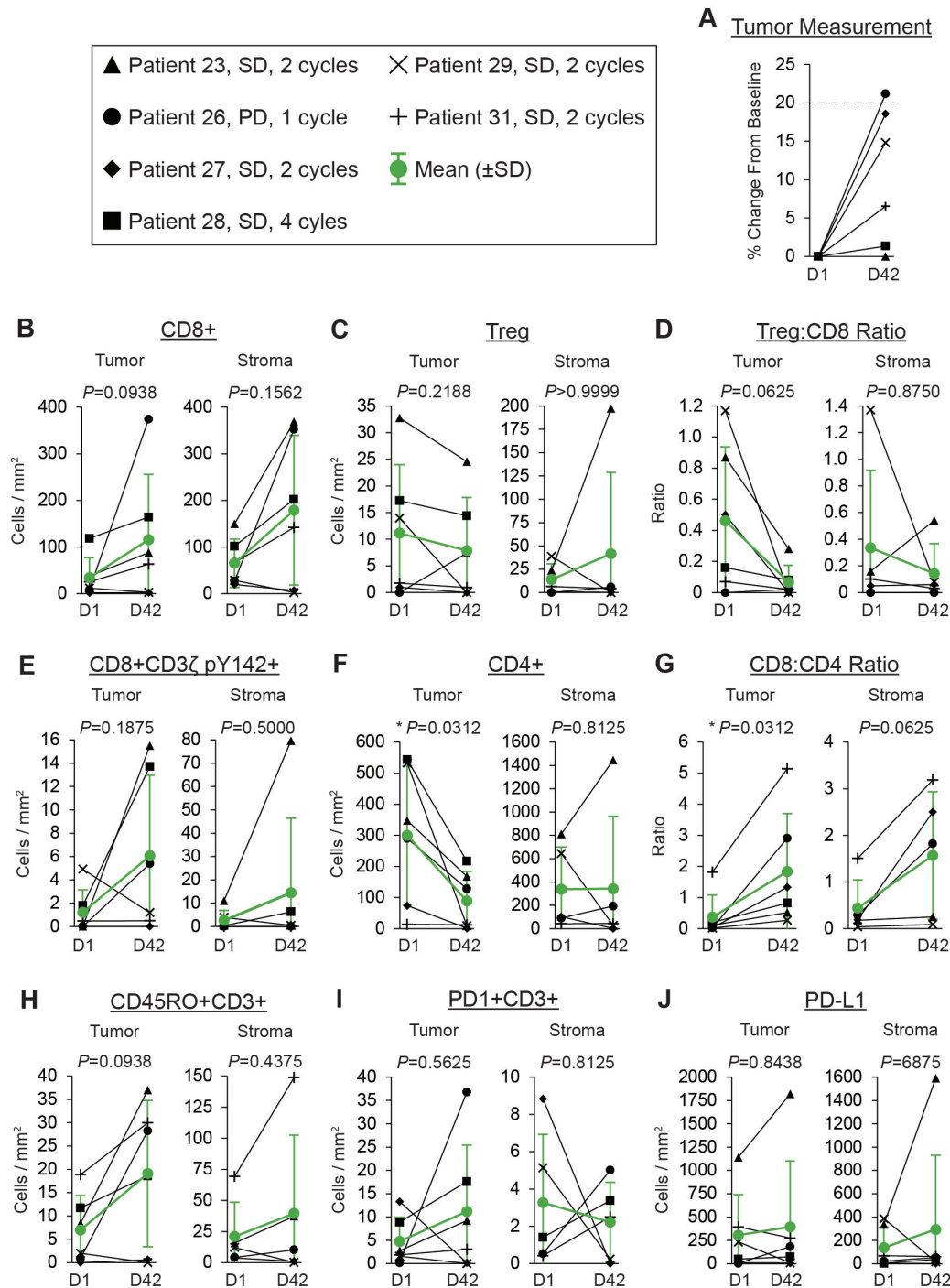


Figure 4 Multiplex immunofluorescence of immune cell populations. Immune cell populations were evaluated in paired biopsies from six patients collected on C1D1 and C1D42. Changes in cell populations are indicated for each patient. P values were calculated for changes in mean values using two-sided Wilcoxon matched-pairs signed-rank tests. (A) Tumor measurements for the six patients are shown for the time points at which biopsies were collected. Minimal changes were measured in patients 23, 28, and 31; patients 27 and 29 nearly reached the threshold for PD and patient 26 did have PD after C1. (B–J) Changes in immune cell populations and ratios between cell populations were measured. Across all analyzed biopsies, significant changes were measured in CD4+cells (decrease, $p=0.0312$, panel F) and CD8+:CD4+ratio (increase, $p=0.0312$, panel G). In addition, changes were measured in individual patients for cell types in which mean changes were not statistically significant. T_{reg} cell populations decreased in both tumor and stromal tissue in biopsies from patient 29 (panel C). Activated CD8+T cell populations expanded in tumors from patients 23 to 28 (activated stromal CD8+T cells increased in biopsies from patient 23 but not from patient 28, panel E) while this population decreased in patient 29 and was not detected at either time point in patient 27. Other cell populations included in this analysis were total CD8+cells (B), memory T cells (H), PD-1+T cells (I), and PD-L1+cells (J). (H) Ratios of T_{reg} cells to total CD8+T cells were also calculated (D). Asterisks indicate p values<0.05. Programmed death 1 receptor, PD-1; programmed death-1; PD-L1, programmed death-ligand 1; SD, stable disease.

T cells) were also measured in these four patients. CD8+CD3 ζ pY142+ T cell (ie, activated CTL) populations increased in tumor tissue from 3/6 patients (patients 23, 26, and 28, all with total CD8+ cell increases). The PD-L1+ cell population expanded in the tumor sections of 3/6 biopsy pairs (patients 23, 26, and 28). T_{reg} cells were observed in tumor tissue from all patients in at least one time point; patients 23, 28, and 31 had T_{reg} cells present in pre-treatment and on-treatment biopsies. T_{reg}:CD8 ratio decreased in 5/6 patients (patients 23, 27, 28, 29, and 31; all with SD); biopsies from patient 26 with PD were negative for T_{reg} cells at baseline but converted to T_{reg}-positive in the on-treatment biopsy. No consistent trends were noted between changes in these cell populations and clinical outcomes.

DISCUSSION

This study was designed to determine the MTD and evaluate the safety profile of the combination of rhIL-15 with nivolumab and ipilimumab. Clinical efficacy and correlative pharmacodynamic analyses were assessed as secondary objectives. While the safety profile of the combination of rhIL-15 with nivolumab and ipilimumab at the MTD was acceptable, clinical benefit was modest. The sample size is relatively small and includes patients with a variety of refractory cancer types to explore efficacy in a variety of tumors.²⁹ The response rate in this study is lower than what has been reported in trials of nivolumab with ipilimumab,³⁰ which may be attributable to the exposure to prior regimens, including immunotherapies, among the study group as well as not selecting for tumors with likelihood of response to immunotherapy (ie, “hot” vs “cold” tumors). The only confirmed PR (cPR) occurred in a patient with cholangiocarcinoma, which is often immunologically “cold” but exhibits features of “hot” tumors in some patients.³¹ This patient was not biopsied, so analysis of immune cell populations in the tumor and TME were not performed to determine whether the regimen induced an objective response in a “cold” tumor. Additionally, the pharmacodynamic analyses were only available on a subset of patients that did not include any PRs or any SD lasting longer than four cycles. PD-L1 expression was not an eligibility criterion. In a study of patients with non-small cell lung NSCLC (ie, an immunologically “hot” tumor type), median overall survival was 17.1 months (95% CI 15.0 to 20.2 months, n=396) versus 17.4 months (95% CI 13.2 to 22.0 months, n=187) among patients with PD-L1 expression of $\geq 1\%$ and $< 1\%$, respectively.³² Despite these limitations, the study provides valuable insights into the safety and pharmacodynamics of rhIL-15 in combination with nivolumab and ipilimumab.

The safety profile of the combination of rhIL-15, nivolumab, and ipilimumab was consistent with the known toxicities of the individual agents or combinations. IL-15 was administered SC based on results from two studies: one in which 2 of 19 patients (11%) experienced serious trAEs following SC administration of

IL-15 at 2 $\mu\text{g}/\text{kg}/\text{day}$ (ie, the highest dose evaluated in this study)¹⁹ and another in which 8 of 27 patients (30%) experienced serious trAEs following intravenous administration of IL-15 at 2 $\mu\text{g}/\text{kg}/\text{day}$.³³ Hypothyroidism or hyperthyroidism, rash, pruritus, pneumonitis, and colitis are among the most prevalent trAEs.³⁴ Here, these AEs were generally manageable with standard treatment algorithms.³⁵ Pneumonitis is a rare but serious and potentially fatal trAE that has been associated with both ipilimumab and nivolumab¹³⁵³⁶; one study reported that pneumonitis was refractory to steroid treatment in 18.5% (12/65) of patients.³⁷ In the present study, one patient at the Triplet 1 dose level experienced grade 1 pneumonitis but recovered and completed an additional seven cycles. In a study of 72 patients with melanoma receiving ipilimumab and nivolumab, 23% of patients experienced colitis of any grade and 17% experienced grade 3/4 colitis.³⁸ Here, colitis was noted in two patients (6%, Doublet A (rhIL-15 with nivolumab) and Triplet 2); both were grade 3 and both patients were removed from the study for colitis. Rash is another prevalent trAE, occurring at any grade in 23% (12 of 52) and at grade 3/4 in 6% (3 of 52) of patients with non-clear cell renal carcinoma receiving ipilimumab with nivolumab.³⁹ Seven of 31 patients (23%) experienced rash or dermatitis on this trial and all were grade ≤ 2 . Hypotension has been reported in 10 of 18 patients (56%) in a study with single-agent IL-15¹⁸ but did not occur in this study. Myocarditis occurred in one patient (Triplet 2); this AE has been reported in 0.27% of patients receiving ipilimumab with nivolumab.⁴⁰ Based on these findings, the addition of rhIL-15 did not increase the frequency or severity of trAEs compared with rates reported in studies of ipilimumab with nivolumab.¹⁶⁴¹

The comprehensive analysis of immune parameters aimed to elucidate the immunological changes induced by the treatment and to inform the development of future combination immunotherapy strategies. Administration of rhIL-15 prior to the administration of nivolumab or ipilimumab led to a transient reduction in NK cell population in blood samples from some patients followed by marked expansion observed on C1D8. These findings are consistent with published reports, although with shorter duration in our study that may result from differences in routes of administration (SC vs IV)³³ or dosages.¹⁸¹⁹ NK cell populations in peripheral blood were modulated by administration of rhIL-15 prior to administration of nivolumab or ipilimumab on C1D8 and decreased between series of rhIL-15 injections. The progressive decrease in magnitude of increase in NK cells with subsequent series may result from tolerance to rhIL-15 administration or could reflect a decrease from a possible peak immediately following the series of rhIL-15 injections. These findings are consistent with the known biological activities of rhIL-15 in stimulating the proliferation and activation of these immune effector cells.¹⁸¹⁹ Blood collection timed specifically to capture these changes (eg, C1D29) may provide more insight on drug effects. Immunological changes were identified following one

cycle of treatment in tumor biopsies from some patients, including the expansion of intratumoral CTLs (CD8+) and activated CTLs (CD8+CD3 ζ pY142+) and the suppression of T_{reg} cells and T_{reg}/CD8 cell ratio; a comparator arm including patients receiving nivolumab and ipilimumab without the addition of rhIL-15 would have been needed to fully characterize the effect of this cytokine. However, the clinical relevance of these findings was not demonstrated in this study. Combining rhIL-15, nivolumab, and ipilimumab increased infiltration and activation of CTLs and NK cells in the tumor microenvironment in preclinical models of colon and prostate cancer.^{22–42} Importantly, increased levels of NK and $\gamma\delta$ T cells in blood did not correlate with increased tumor cell infiltration at C1D42 in this study. Lymphopenia occurred in 7 of 26 patients (27%) in the Triplet cohort, a rate consistent with those reported from other immune checkpoint inhibitor studies.^{43–44} The correlation between lymphopenia and poor prognoses supports the rationale for combining checkpoint inhibitors with immunostimulatory cytokines such as IL-15.⁴⁵ However, the variable changes in PD-L1 expression suggest a complex interaction between the immune system and the tumor, warranting further investigation.⁴⁶

Combining rhIL-15 with checkpoint inhibitors has been suggested as a promising therapeutic strategy for patients with cancers.^{17–47–48} The safety of this combination observed in this study was acceptable. However, given that the only cPR and the longest duration of SD occurred in patients receiving the lowest dose of rhIL-15 (ie, Triplet 1), future studies may not require dosing at the MTD and could focus on refining the dosing schedule and route of administration,³³ IL-15 formulation,⁴⁹ or combinations with other therapeutic modalities⁵⁰ to maximize the therapeutic potential of this novel immunotherapeutic approach.^{51–52} While dose-proportional increases in serum cytokine levels (eg, IL-1B, IL-6, and tumor necrosis factor- α (TNF α)) have been measured in response to IL-15 administration, no relationship between serum cytokine levels and clinical response has been reported.³³ While cytokine levels were not measured in our study, our findings are consistent with immune modulation by the combination of rhIL-15 with nivolumab and ipilimumab without observed clinical response. Based on these data, further analyses exploring the impact of negative regulators of cytokine signaling, such as suppressor of cytokine signaling (SOCS) proteins, as well as other immune checkpoint molecules and the influence of the TME, are warranted.^{51–52}

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